MIL-P-60356A (MU)
10 March 1971
SUPERSEDES
MIL-P-60356 (MU)
22 April 1966

MILITARY SPECIFICATION

POWDER, IGNITER, CLEAN BURNING (FOR USE IN CHARGES, PROPELLING, M3A1 M4A2 and XM119E4 FOR 155MM HOWITZER)

1. SCOPE

1.1 This specification covers two types of powder, clean burning igniter (CBI) for use in Charges, Propelling, M3A1 and M4A2 for 155MM Howitzer. (see 6.1)

1.2 Classification.-The CBI shall be of the following types:

Type I - Solid porous flakes
Type II - Perforated non-porous flakes

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-D-98	-	Diphenylamine
MIL-G-155	-	Graphite
MIL-P-156	-	Potassium Nitrate
MIL-C-206	-	Cellulose, Cotton
MIL-C-216	-	Cellulose, Woodpulp, Sulfite
MIL-N-244	-	Nitrocellulose (For use in explosives)

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-286	-	Propellant, Solid: Sampling, Examination, and Testing
MIL-STD-652	-	Propellants, Solid for Cannons: Requirements and Packing

FSC: 1376

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:30Z
Check the source to verify that this is the current version before use.

ue. the Government inspector ... ments as a sub

DRAWINGS

ARMY

9207624 - Charge, Propelling, 155MM: M4A2, M1, M45, and M126

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR-171-190 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Rules are now a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Officer, Washington, D.C., 20402. Orders for the above publications should cite "49 CFR 171-190 (latest revision)".

3. REQUIREMENTS

3.1 Materials.-The raw materials used in the manufacture of the igniter composition shall be in accordance with requirements of the following applicable specifications:

SPECIFICATION	TYPE	GRADE	CLASS
MIL-D-98	- - -	- - -	- - -
MIL-G-155	- - -	III or IV	- - -
MIL-P-156	- - -	- - -	2
MIL-N-244	1	C	- - -

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:30Z
Check the source to verify that this is the current version before use.

3.1.1 Cellulose.-The cellulose used to manufacture the nitrocellulose shall be made with the following:

Type I - use MIL-C-206
Type II - use MIL-C-216

3.1.2 Chemical and Physical properties shall comply with the following when tested as specified in 4.3.1:

MATERIAL	PERCENTAGE	
TYPE	I	II
Nitrocellulose, min.	98.0	98.0
Diphenylamine, ± 0.20	1.50	1.50
Potassium Nitrate (added)	0.1 maximum	-----
Graphite (glaze added)	0.2	0.3
Physical Properties		
Volatiles, ± 0.5	1.3	1.3
Bulk Density, lbs/ft^3	32.0 ± 5.0	27.0 ± 3.0
Ash, max	0.4	0.4

3.2 Grain dimension.-The CBI powder shall be in the form of flakes and shall have the following ADVISORY measurements when tested as specified in 4.3.2.

TYPE	I	II
Diameter, inch	.028	0.055
Thickness, web, inch	.005	0.004
Perforation, dia. inch	- - -	0.010

3.3 Heat test.-The composition when subjected to a 134.5 degree Centigrade (°C) heat test shall not change the test paper to a salmon pink color in less than 50 minutes and shall not explode in less than 5 hours when tested as specified in 4.3.6.

3.4 Appearance.-CBI powder shall be uniformly grained and free from foreign materials.

3.5 Ballistics.-Clean burning Igniter (CBI) when fired from a M114A1, 155MM, Howitzer under standard conditions with an M4A2 Propelling Charge at Zone VII and a 95.0 lb. M107 Projectile shall comply with the following requirements.

3.5.1 Uniformity of Muzzle Velocity.-The standard deviation of velocity shall not exceed 7.2 fps at +70°F or -65°F when tested in accordance with 4.3.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:30Z
Check the source to verify that this is the current version before use.

3.5.2 Functioning.-The CBI shall function satisfactorily without misfires or hangfires when tested in accordance with 4.3.7.

3.6 Workmanship.-The best commercial practices shall be used in the formulation of the composition furnished under this specification, and all other applicable documents. The composition and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground nitrocellulose shall be used in the manufacture of the composition procured under this specification unless otherwise specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.

b. Number of batches of product inspected, (shall be made available upon request by the Contracting Officer).

c. Results obtained, by defect code, for all inspections performed, (shall be made available upon request by the Contracting Officer).

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of pounds in the lot.

g. Date submitted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:30Z
Check the source to verify that this is the current version before use.

h. Eight copies of propellant description sheet SMU Form 1047 containing the applicable data (see 6.5). The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-A lot shall consist of igniter composition produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch will consist of that quantity of igniter composition that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

4.2.2 Testing

4.2.2.1 Selection and submission of Proving Ground Sample.

4.2.2.1.1 Initial Production Lots.-Beginning with the first lot and continuing until three (3) successive lots have complied with the requirements of 3.5, twenty (20) pounds of CBI powder shall be forwarded to a Government Proving Ground as designated by the contracting officer. Selection, of the sample shall be as follows:

a. Select ten (10) containers at random from the lot (if lot consists of less than 10 containers select all containers
b. Extract an equal (approximate) amount of powder from each container with a scoop to achieve the total sample weight
c. Place powder removed from each container in an individual package and ship to proving ground.

4.2.2.1.2 Subsequent Production.-After three (3) consecutive lots of a new interfix have complied with 3.5, the CBI sample size shall be reduced to 2 pounds (lbs). Selection and submission shall be as stated (except for sample size) in 4.2.2.1.1.

4.2.2.2 Sampling for chemical and physical test.-Sampling for chemical and physical test shall be in accordance with paragraph 4.4.2 of Standard MIL-STD-652.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:30Z
Check the source to verify that this is the currentversion before use.

4.2.2.2.1 Ballistics

4.2.2.2.2 Initial Production Lots.-Beginning with the first lot and continuing until three (3) successive lots have complied with the requirements of 3.5. Ballistic sampling and acceptance shall be in accordance with Table I below.

TABLE I

Temperature	Sample Size	Characteristic	Acceptance Criteria
70 degrees F	25	Velocity Std. Deviation	Reject if std. dev. of observed velocities X factor 0.81 exceeds 7.2 fps (major defect)
		Misfires	Reject if one or more misfires attributable to CBI powder are experienced (major defect)
		Hangfires	Refer if one or more audible hangfires are experienced or one or more ignition delay times greater than 500 milliseconds (ms) is recorded
-65 degrees F	25	Same as 70 degrees firing	Same as 70 degrees F firings

4.2.2.2.3 Subsequent lots.-After three (3) successive lots of a new interfix have satisfied the requirements of 3.5, Ballistic Sampling and acceptance shall be in accordance with Table II.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:30Z
Check the source to verify that this is the currentversion before use.

TABLE II

Temperature	Sample Size	Characteristic	Acceptance Criteria
70 degrees F	5	Misfire	Reject if one or more misfires attributable to CBI powder are experienced (major defect) (see 3.5.2)
		Hangfires	Refer if one or more audible hangfires are experienced during testing or if one or more ignition delay times greater than 500 ms is recorded (see 3.5.2)

4.3 Test Methods and Procedures

4.3.1 Determination of composition (see 6.3)

4.3.1.1 Nitrocellulose.-Nitrocellulose content shall be determined in accordance with Method 209.2 of MIL-STD-286, Code No. 01001.

4.3.1.2 Potassium Nitrate.-Potassium Nitrate content shall be determined in accordance with Method 310.4 of Standard MIL-STD-286, Code No. 01002.

4.3.1.3 Diphenylamine.-Diphenylamine content shall be determined in accordance with Method 201.1 of Standard MIL-STD-286, Code No. 01003.

4.3.1.4 Graphite.-Graphite content shall be determined in accordance with Method 309.1 of Standard MIL-STD-286, Code No. 01004.

4.3.2 Grain dimension.-The grain dimension shall be determined in accordance with Method 504.1 of Standard MIL-STD-286, Code No. 02001.

4.3.3 Volatiles.-The total volatiles content shall be determined in accordance with Method 103.3 of Standard MIL-STD-286, Code No. 03001.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:30Z
Check the source to verify that this is the current version before use.

4.3.4 Bulk density.-The bulk density shall be determined in accordance with Method 502.1.1 or 502.2.1 of Standard MIL-STD-286, Code No. 04001.

4.3.5 Ash.-The ash content shall be determined in accordance with Method 106.2 of Standard MIL-STD-286, Code No. 05001.

4.3.6 Heat test.-Heat tests shall be conducted in accordance with Method 404.1 of Standard MIL-STD-286, Code No. 06001.

4.3.7 Ballistics.-All testing shall be conducted at a Government Proving Ground in accordance with applicable Acceptance Test Procedure, Code No. 07001.

4.3.7.1 Initial Production Lots.-Fifty igniter bag charges shall be assembled with sample powder submitted by the manufacturer taking care that one bag charge contains powder from not more than one package (see 4.2.2.1). Twenty-five of these igniter bag charges shall be assembled into M4A2 Propelling Charges, conditioned to -65 degrees F (with a mylar protective bag) and fired with an M107 Projectile within 3 minutes after removal from conditioning box. The remaining twenty-five shall be conditioned to 70 degrees F and similarly fired. The propelling charge shall be positioned a minimum of five inches from the rear face of the tube prior to firing. A MK2A4 or M82 Primer manufactured within five years of date of testing shall be used. These Primers shall not be conditioned. All propelling charges shall be fired at Zone VII and the igniter protective cap shall not be removed until just prior to insertion of the propelling charge into the weapon. A minimum of 10 control rounds loaded with reference CBI powder shall be fired in conjunction with the test items. Velocity, and ignition delay times shall be electronically recorded and observation for misfires and hangfires shall be made.

4.3.7.2 Subsequent lots.-Testing shall be conducted as specified in 4.3.7.1 with the exception that only five igniter bag charges are required for both test (70 degrees F only) and control samples. Also muzzle velocity measurements are not needed.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Standard MIL-STD-652.

6. NOTES

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:30Z
Check the source to verify that this is the currentversion before use.

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Packing required (see 5.1).
c. Process.-Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 Process control.-The data obtained from the composition chemical analysis for each batch of igniter composition for the first five lots shall be forwarded to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2, Dover, New Jersey 07801.

6.3 Inspection Code Numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.4 Intended use.-It is intended that the item of this Specification will be used in 155MM Howitzer.

6.5 Submission of SMU Form 1047.-Information contained in propellant description sheet, SMU Form 1047, need not be resubmitted in the form of a certificate provided the provisions regarding certifying of responsible agents is adhered to (see 4.1.1).

6.6 Standard deviation.-The standard deviation should be calculated with (N-1) as the division in a standard statistical technique equivalent to that shown in Standard MIL-STD-414, Section B, Example B-1.

6.7 Chemical stability testing.-Five pounds of igniter composition shall be selected at random from the lot, and shall be forwarded to the Commanding Officer, Picatinny Arsenal, Dover, New Jersey for the 65.5 degree Centigrade surveillance test in accordance with Standard MIL-STD-286, Method T407.1.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project Number: 1376-A-070

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:30Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:30Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:30Z
Check the source to verify that this is the currentversion before use.

FOLD

DEPARTMENT OF THE ARMY
PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

COMMANDING OFFICER
United States Army
Picatinny Arsenal
ATTN: SMUPA-ND
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:30Z
Check the source to verify that this is the current version before use.